

Extremal exponent one for consecutive log-concavity failures in subcubic trees

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Abstract

Let $M_\Delta(N)$ be the maximum length of a consecutive run of strict log-concavity failures among independence polynomials of trees of order at most N and maximum degree at most Δ . We give a self-contained construction proving $\log M_3(N)/\log N \rightarrow 1$. The construction uses binary branching and terminal gadgets whose limiting root odds grow exponentially in their order. For each fixed $\varepsilon > 0$, a gadget of order $O(\log(1/\varepsilon))$ suffices to obtain $\Omega_\varepsilon(N^{1-\varepsilon})$ consecutive failures. We prove the finite-height transfer through a uniform local limit theorem with two derivatives. A separate, fully specified certificate for a 14-vertex gadget proves at least $2^{-86}N^{47/50}$ consecutive failures at every height $h \geq 1024$, below 0.99α . Degree two permits no failures. A finite covariance argument also gives $M_\Delta(N) = o(N)$ for every fixed degree bound, so exponent one does not mean positive linear density.

Prior work and verification scope. The exponent-one conclusion for subcubic trees is already reported in Christopher’s preprint of 12 September 2026 [1]. It is therefore *not claimed here as a first discovery*. The present text supplies its own compressed-gadget construction, proof, and finite-height certificate. The general theorem has an analytic proof with constants depending on the selected gadget. The explicit 14-vertex theorem has a specified interval, starting height and numerical error bounds. The executed verifier checks those bounds using rigorous ball arithmetic; it is not a proof-assistant formalization or independent peer review. The literature statement about [1] is based on its indexed abstract; that manuscript’s full proof was not audited here.

1 The exact extremal statement

Write $i_k(T)$ for the number of independent sets of size k in T , and $\alpha(T)$ for its independence number. A strict failure at k is

$$i_k(T)^2 < i_{k-1}(T)i_{k+1}(T), \quad 1 \leq k < \alpha(T).$$

A run consists of consecutive failure indices. Define

$$M_\Delta(N) = \max_{\substack{T \text{ a tree, } |V(T)| \leq N \\ \Delta(T) \leq \Delta}} \{\text{length of a run in } T\},$$

with value zero when no run exists.

Theorem 1. *For every fixed integer $\Delta \geq 3$,*

$$\boxed{\lim_{N \rightarrow \infty} \frac{\log M_\Delta(N)}{\log N} = 1.} \tag{1}$$

More explicitly, for every $0 < \varepsilon < 1$ there exist $c_\varepsilon > 0$ and $N_\varepsilon < \infty$ such that

$$M_3(N) \geq c_\varepsilon N^{1-\varepsilon} \quad (N \geq N_\varepsilon).$$

For $\Delta \leq 2$, $M_\Delta(N) = 0$ for all N .

The quantifiers matter: one fixes ε and one terminal gadget, then lets the binary height grow. No uniformity in ε is assumed when proving (1). In particular, the theorem asserts $N^{1-o(1)}$, not $\Omega(N)$. Section 7 proves separately that $M_\Delta(N)/N \rightarrow 0$ for every fixed Δ .

2 An exponentially efficient terminal gadget

Let G_0 be an edge rooted at an endpoint. To obtain G_{t+1} , take one new root v , its single child w , and make the roots of two disjoint copies of G_t the children of w . Thus

$$n_t := |V(G_t)| = 2^{t+2} - 2, \quad s_t = 2^{t+1} - 1 = n_t/2. \quad (2)$$

The maximum degree is three, and the root has degree one. Both root-conditioned independence numbers equal s_t . Indeed, if both conditional maxima of G_t are s_t , the state with w occupied has size $1 + 2s_t$, and the state with v occupied has the same size. The alternative with both v, w absent has size $2s_t$. This proves the assertion by induction.

Let $R_t(\lambda)$ denote the occupied/unoccupied partition-function ratio at the root. The construction gives the exact recursion

$$R_0(\lambda) = \frac{\lambda}{1+\lambda}, \quad R_{t+1}(\lambda) = \frac{\lambda(1+R_t(\lambda))^2}{\lambda + (1+R_t(\lambda))^2}. \quad (3)$$

Define integers $A_0 = 1$, $A_{t+1} = (1 + A_t)^2$. Their first values are 1, 4, 25, 676, 458329. For $t \geq 1$,

$$A_t \geq 2^{2^t}. \quad (4)$$

Lemma 2. *For every $t \geq 1$ there is $r_t \in (A_t - 1, A_t)$ such that*

$$\lambda_t = r_t(1 + r_t)^2, \quad R_t(\lambda_t) = r_t. \quad (5)$$

Moreover, for every $\lambda > 0$,

$$0 < R_t(\lambda) < A_t, \quad A_t - R_t(\lambda) \leq \frac{2A_t^2}{\lambda}, \quad 0 < \frac{d \log R_t}{d \log \lambda} \leq \frac{2A_t}{\lambda}. \quad (6)$$

Proof. The first two claims hold at $t = 0$. Put $a = A_t$, $R = R_t$, and $S = (1 + R)^2$. Then

$$(1 + a)^2 - R_{t+1} \leq 2(1 + a)(a - R) + \frac{(1 + a)^4}{\lambda} \leq \frac{4(1 + a)a^2 + (1 + a)^4}{\lambda} \leq \frac{2(1 + a)^4}{\lambda}.$$

The last inequality is $4a^2 \leq (1 + a)^3$, valid for $a \geq 1$ since $(1 + a)^3 - 4a^2 = a^2(a - 1) + 3a + 1 > 0$. For $d_t = d \log R_t / d \log \lambda$ and $e = S / (\lambda + S)$, differentiation gives

$$d_{t+1} = e + 2(1 - e) \frac{R}{1 + R} d_t \leq \frac{A_{t+1}}{\lambda} + 2d_t \leq \frac{A_{t+1} + 4A_t}{\lambda} \leq \frac{2A_{t+1}}{\lambda}.$$

Positivity is preserved, and the base is $d_0 = 1/(1 + \lambda)$. Here $A_{t+1} \geq 4A_t$ follows from $(A_t - 1)^2 \geq 0$.

For $A = A_t \geq 4$, set $r = A - 1$, so $\lambda = (A - 1)A^2$. The second bound gives $R_t(\lambda) \geq A - 2/(A - 1) > A - 1$. At $r = A$, the first bound gives $R_t(r(1 + r)^2) < r$. Continuity gives a root between these two values. Uniqueness is not needed: choose any root, and keep it fixed. $\square$

3 Binary amplification and its smoothing limit

Fix $t \geq 4$ for the general proof. Write $s = s_t$, $A = A_t$, and choose a root of (5). Suppress t from the activity and odds. Let $T_0 = G_t$, and form T_{h+1} by attaching the roots of two copies of T_h to one new root. Its degree is at most three, and

$$N_h = (2s + 1)2^h - 1, \quad \alpha_h = s2^h + \sum_{j=0}^{\lfloor (h-1)/2 \rfloor} 2^{h-1-2j}. \quad (7)$$

The second formula follows from the two root-conditioned maximum recurrences, starting from equal maxima s . In particular $\alpha_h/2^h \rightarrow s + 2/3$. At the chosen activity every root has odds r , because the next odds are $\lambda/(1+r)^2 = r$. Put

$$q = \frac{r}{1+r}, \quad p = 1 - q, \quad \rho = 2q, \quad \ell = \frac{\sqrt{2}}{\rho} < \frac{3}{4}, \quad \theta = \frac{1}{1+\rho}. \quad (8)$$

Here $p < 1/A \leq 1/458329 < 1/2048$ and $q > 99/100$.

Let X_h be the independent-set size under the activity λ . Write $\mu_h^a = \mathbb{E}(X_h \mid \text{root} = a)$ and $D_h = \mu_h^1 - \mu_h^0$. The seed derivative in (6) is precisely D_0 . Thus

$$0 < D_0 < \frac{2}{A(A-1)}, \quad D_{h+1} = 1 - \rho D_h.$$

Consequently, with $a_0 = D_0 - \theta$,

$$D_h = \theta + a_h, \quad a_h = a_0(-\rho)^h, \quad \frac{1}{4} < |a_0| < \frac{1}{2}, \quad \theta/|a_0| < 2. \quad (9)$$

In particular the oscillating scale does not vanish.

Set $Y_h^a = (X_h - \mu_h^a)/a_h$ conditionally on the root state, and $z_h = D_h/a_h$. Independence of the child subtrees gives

$$\begin{aligned} Y_{h+1}^0 &\stackrel{d}{=} -\rho^{-1} \sum_{j=1}^2 \{Y_{h,j}^{S_j} + (S_j - q)z_h\}, \\ Y_{h+1}^1 &\stackrel{d}{=} -\rho^{-1} \sum_{j=1}^2 Y_{h,j}^0, \end{aligned} \quad (10)$$

where S_j are independent Bernoulli variables of parameter q , and the conditional child copies are independent.

Lemma 3. *The transformation (10) with z_h replaced by one has a unique fixed pair of centered finite-variance laws (Y^0, Y^1) . If $C = 2s + 5$, then*

$$\max_a W_2(Y_h^a, Y^a) \leq C\ell^h, \quad W_2(Z_h, Z) \leq (C+1)\ell^h, \quad (11)$$

where $m_h = \mathbb{E}X_h$, $Z_h = (X_h - m_h)/a_h$, and $Z = Y^S + (S - q)$. For $\nu = 2/\rho^2$, the conditional variances satisfy

$$v_1 = \nu v_0, \quad v_0 = \frac{\nu qp}{1 - \nu p - q\nu^2}, \quad \frac{p}{5} < v_a < p. \quad (12)$$

Both conditional laws are subgaussian with a common variance proxy

$$\sigma^2 = \frac{\nu}{4(1-\nu)} < \frac{1}{3}. \quad (13)$$

Proof. Couple two centered pairs and use independent child couplings. The cross terms vanish by centering, so the maximum of their two W_2 distances contracts by $\sqrt{2}/\rho = \ell$. Completeness gives the fixed pair. Taking variances gives (12). For its bounds, $1/2 < \nu < \frac{1}{2}(100/99)^2$, and

$$1 - \nu p - q\nu^2 > 1 - \frac{1}{200}(100/99)^2 - \frac{1}{4}(100/99)^4 > 7/10 > \nu.$$

Thus $v_0 < p$, $v_1 < v_0$. Conversely, the denominator is less than one, so $v_0 > qp/2$ and $v_1 > qp/4 > p/5$.

A common subgaussian proxy u is mapped to $\nu(u + 1/4)$: condition on S , apply the conditional bound, and then the centered Bernoulli bound. Iteration from point masses and Fatou's lemma

give (13). Each initial conditional size lies in $[0, s]$, so its normalized standard deviation is at most $2s$ by (9). The initial W_2 error is therefore at most $2s + 1$. The perturbation $z_h - 1 = \theta/a_h$ contributes at most $\ell\rho^{-h}$ at the next step. Summing gives

$$\max_a W_2(Y_h^a, Y^a) \leq \ell^h \left(2s + 1 + \sum_{j \geq 0} 2^{-j/2} \right) < (2s + 5)\ell^h.$$

The unconditional mixture adds at most $\rho^{-h} \leq \ell^h$. $\square$

4 Uniform finite-height inversion through two derivatives

The following step is necessary: convergence of four moments, or weak convergence alone, would not imply a consecutive coefficient run. Let $\psi_a(u) = \mathbb{E}e^{iuY^a}$ and $\phi(u) = \mathbb{E}e^{iuZ}$. Then

$$\psi_0(u) = \phi(-u/\rho)^2, \quad \psi_1(u) = \psi_0(-u/\rho)^2, \quad \phi(u) = pe^{-iqu}\psi_0(u) + qe^{ipu}\psi_1(u). \quad (14)$$

For $Q(u) = \max_a |\psi_a(u)|$, this implies $Q(u) \leq Q(u/\rho)^2$. Define the explicit constants

$$\tau = p/16, \quad \eta = \tau/\rho, \quad c = p^3/2^{15}, \quad \gamma = \log_\rho 2 > 1. \quad (15)$$

Lemma 4. For $|u| \geq \eta$,

$$Q(u) \leq \exp[-c(|u|/\tau)^\gamma]. \quad (16)$$

The law of Z has a strictly positive entire density f . If ϕ_h is the characteristic function of Z_h , then

$$g_h(x) = \frac{1}{2\pi} \int_{-\pi|a_h|}^{\pi|a_h|} e^{-iux} \phi_h(u) du \quad (17)$$

satisfies $\|g_h^{(j)} - f^{(j)}\|_\infty \rightarrow 0$ for $j = 0, 1, 2$. At every lattice node $x_{h,k} = (k - m_h)/a_h$,

$$g_h(x_{h,k}) = |a_h| \mathbb{P}(X_h = k). \quad (18)$$

Proof. For an independent copy Y' of Y^a , the subgaussian bound gives $\mathbb{E}(Y^a - Y')^4 \leq 64\sigma^4 < 8$. The inequality $\cos x \leq 1 - x^2/2 + x^4/24$ yields, for $|u| \leq \tau$,

$$|\psi_a(u)|^2 \leq 1 - v_a u^2 + u^4/3 \leq 1 - \frac{1}{2}v_a u^2.$$

Indeed $u^2 \leq p^2/256 < (3/2)v_a$. Therefore $Q(u) \leq e^{-pu^2/20}$. On $[\eta, \tau]$, since $\eta > p/32$, this is at most e^{-c} . Iterating $Q(u) \leq Q(u/\rho)^2$ proves (16). The same bound applies to $|\phi|$. Because $\gamma > 1$, Fourier inversion defines an entire density f . It is nonnegative and not identically zero, hence positive almost everywhere. The first equation of (14) expresses the law of Y^0 as a scaled convolution of two copies of f , which is strictly positive at every real argument. The mixture then gives $f > 0$ everywhere.

For the finite-height estimate use physical frequencies and write $Q_h(v) = \max_a |\mathbb{E}(e^{ivX_h} | a)|$. Root conditioning gives

$$Q_h(v) \leq Q_k(v)^{2^{h-k}} \quad (0 \leq k \leq h). \quad (19)$$

Choose

$$h_0 = \max \left\{ 1, \left\lceil \frac{\log(4C\tau/c)}{\log(1/\ell)} \right\rceil \right\}.$$

By (11), for every $k \geq h_0$ and $\eta \leq |u| \leq \tau$,

$$Q_k(u/a_k) \leq e^{-c} + C\tau\ell^k \leq e^{-c} + c/4 \leq e^{-c/4}.$$

For $|u| \geq \eta$ and $|u/a_h| \leq \eta/|a_{h_0}|$, select $k \in [h_0, h]$ with $\eta \leq |a_k u/a_h| \leq \tau$. Applying (19) then gives

$$|\phi_h(u)| \leq \exp[-(c/4)(|u|/\tau)^\gamma]. \quad (20)$$

There is a separate bound for the remaining physical frequencies. In each conditional seed law, two adjacent sizes have probability at least $(1 + \lambda)^{-n_t}$. For root zero use the empty set and a singleton; for root one use the root alone and the root plus a vertex at distance two. Such a vertex exists for $t \geq 1$. For adjacent masses a, b , the elementary bound $|\chi(v)|^2 \leq 1 - 2ab(1 - \cos v)$ gives

$$Q_0(v) \leq e^{-d_t v^2} \quad (|v| \leq \pi), \quad d_t = \frac{1}{8\{1 + A_t(1 + A_t)^2\}^{2n_t}} > 0.$$

We used $1 - \cos v \geq 2v^2/\pi^2$ and $\lambda < A_t(1 + A_t)^2$. Thus for $|u/a_h| \geq \eta/|a_{h_0}|$ and $|u| \leq \pi|a_h|$,

$$|\phi_h(u)| \leq \exp[-d_t \eta^2 |a_{h_0}|^{-2} 2^h]. \quad (21)$$

The integration range grows exponentially in h , while this bound decays exponentially in 2^h . On the complementary range, (20) is a common integrable bound even after multiplication by $|u|^2$. On each fixed compact frequency interval, (11) gives uniform convergence. Split the inverse transforms into compact, low-physical-frequency tail, and high-physical-frequency tail parts; the three estimates prove convergence in C^2 , uniformly in x . Finally (18) is the ordinary lattice Fourier inversion formula, with the change of variable $u = a_h v$. $\square$

5 A valley, a curvature interval, and the exponent

Consider three intervals of radius $1/4$, with centers $-q$, $1/2 - q$, and $1 - q$. Denote their probabilities under Z by L, V, R . By Chebyshev's inequality and (12),

$$L \geq p(1 - 16v_0) > p/2, \quad R \geq q(1 - 16v_1) > 1/2, \quad V \leq 16(pv_0 + qv_1) < 16p. \quad (22)$$

Since $p < 1/2048$, it follows that $V^2 < LR$. If f were log-concave on the union of the intervals, its restriction extended by zero would be log-concave. Integrating over translates of a fixed interval preserves log-concavity, by the Prékopa–Leindler inequality, and would imply $V^2 \geq LR$. This contradiction proves that f is not log-concave there.

Since f is positive and smooth, there exist a nondegenerate closed interval $J_t \subset [-5/4, 5/4]$ and $\zeta_t > 0$ such that

$$\inf_{J_t} f > 0, \quad (\log f)'' \geq 2\zeta_t \quad \text{on } J_t. \quad (23)$$

The C^2 convergence proves that, for all sufficiently large h , $g_h > 0$ and $(\log g_h)'' \geq \zeta_t$ throughout J_t . All integers k whose three nodes $x_{h,k-1}, x_{h,k}, x_{h,k+1}$ lie in J_t form a consecutive interval of size at least $|J_t||a_h| - 3 = \Omega_t(\rho^h)$. Twice integrating the curvature with the triangular kernel gives

$$\log \frac{g_h(x_{h,k-1})g_h(x_{h,k+1})}{g_h(x_{h,k})^2} \geq \frac{\zeta_t}{|a_h|^2} > 0.$$

The activity cancels exactly:

$$\frac{\mathbb{P}(X_h = k-1)\mathbb{P}(X_h = k+1)}{\mathbb{P}(X_h = k)^2} = \frac{i_{k-1}(T_h)i_{k+1}(T_h)}{i_k(T_h)^2}. \quad (24)$$

Thus every index in this interval is a strict failure.

For completeness the nodes are inside the coefficient range. Put $\pi_* = q/(1 + q)$ and $b_j = \pi_* + (q - \pi_*)(-q)^j$. The occupation probability at distance j from the top is b_j , so

$$m_h = 2^h(\mu_0^0 + b_h D_0) + \sum_{j=0}^{h-1} 2^j b_j, \quad \frac{m_h}{2^h} \longrightarrow L_t := \mu_0^0 + \pi_* D_0 + \pi_*. \quad (25)$$

The seed mixture has mean at most s , giving $0 < L_t < s + 1/2 < s + 2/3$. Since $a_h/2^h = a_0(-q)^h \rightarrow 0$, all the selected nodes lie strictly between zero and α_h at large height. They even remain a fixed, t -dependent proportional distance below α_h .

Completion of the extremal proof. Let $\kappa_t = \log_2 \rho_t$. From $r_t > A_t - 1$,

$$\kappa_t = 1 - \frac{\log(1 + 1/r_t)}{\log 2} > 1 - \frac{1}{(A_t - 1) \log 2} \rightarrow 1. \quad (26)$$

The run above has length $\Omega_t(N_h^{\kappa_t})$. For a given $\varepsilon > 0$, select a fixed $t \geq 4$ with $\kappa_t > 1 - \varepsilon$. For arbitrary sufficiently large N , take the largest h for which $N_h \leq N$. Since $N_{h+1} = 2N_h + 1$, $N_h > (N - 1)/2$, which transfers the same power lower bound to every large N , not merely a subsequence. Therefore

$$\liminf_{N \rightarrow \infty} \frac{\log M_3(N)}{\log N} \geq 1 - \varepsilon.$$

Let $\varepsilon \downarrow 0$. The trivial upper bound $M_\Delta(N) \leq N$ and the inclusion $M_\Delta \geq M_3$ for $\Delta \geq 3$ prove (1).

Trees of maximum degree at most two are paths. Their coefficients are $i_k(P_n) = \binom{n-k+1}{k}$, and

$$\frac{i_{k+1}(P_n)}{i_k(P_n)} = \frac{(n-2k+1)(n-2k)}{(k+1)(n-k+1)}$$

is decreasing over its valid indices: the numerator decreases and the denominator increases there. Thus these sequences are log-concave, proving the degree-two assertion and completing Theorem 1. $\square$

Size of the terminal gadget. By (4), it suffices to take

$$t = \max \{4, \lceil \log_2 \log_2(2 + 2/\varepsilon) \rceil \}.$$

Then $A_t \geq 2 + 2/\varepsilon$, so (26) gives $\kappa_t > 1 - \varepsilon$. Formula (2) yields

$$n_t \leq 64 + 8 \log_2(2 + 2/\varepsilon).$$

This bounds the terminal size, not the starting height or the coefficient in the run bound uniformly in ε .

6 A fully specified instance: fourteen terminal vertices

The general argument used $t \geq 4$ for a simple analytic valley bound. A separate validated computation works already at $t = 2$, where $s = 7$ and $A_2 = 25$. Choose the root $r \in (24, 25)$ of (5) by the rational bisection implemented in the verifier, and define $\lambda, q, \rho, a_h, m_h$ as above. Numerically, for orientation only,

$$r \approx 24.9509982314, \quad \lambda \approx 16803.357278, \quad \rho \approx 1.92293167368.$$

Let $x_* = -2/5$, $d = 2^{-80}$, $J = [x_* - d, x_* + d]$, and

$$I_h = \{k \in \mathbb{Z} : x_{h,k-1}, x_{h,k}, x_{h,k+1} \in J\}.$$

Theorem 5. *For every integer $h \geq 1024$, the unweighted subcubic tree T_h with seed G_2 has order $N_h = 15 \cdot 2^h - 1$ and satisfies*

$$|I_h| \geq 2^{-82} \rho^h \geq 2^{-86} N_h^{47/50}, \quad (27)$$

$$0 < k < \frac{99}{100} \alpha_h \quad (k \in I_h), \quad (28)$$

$$\log \frac{i_{k-1}(T_h) i_{k+1}(T_h)}{i_k(T_h)^2} \geq \frac{400}{|a_h|^2} > 0 \quad (k \in I_h). \quad (29)$$

6.1 Initial data and the limiting characteristic functions

The file `verify_subcubic.py` uses Python-FLINT 0.9.0 at 512-bit precision. All sign decisions use exact rational arithmetic or rigorous Arb/Acb ball enclosures. The root interval is obtained by 400 exact rational bisections of $R_2(r(1+r)^2) - r$ on $[24, 25]$. The endpoint signs are checked before and after bisection, so the interval contains an actual root. For conditional seed polynomials P_0, P_1 the exact update is

$$(P_0, P_1) = (1 + \lambda, \lambda) \quad \text{initially,} \quad (P_0, P_1) \mapsto ((P_0 + P_1)^2 + \lambda P_0^2, \lambda(P_0 + P_1)^2).$$

Two updates give G_2 . Evaluating deficiency moments from these polynomials certifies

$$\begin{aligned} 2^{14} < \lambda < 2^{15}, \quad 24/25 < q < 25/26, \quad 48/25 < \rho < 2, \\ 0 < D_0 < 1/100, \quad 1/4 < |a_0| < 1, \quad \theta/|a_0| < 2, \\ \mu_0^0 > 6, \quad \text{Var}(X_0 | a)/a_0^2 < 1, \quad v_a < 1/16, \quad \sigma^2 < 1/3. \end{aligned} \quad (30)$$

Here the variance formulas and contraction argument remain valid because $q > 24/25$. Their explicit specialization is

$$\max_a W_2(Y_h^a, Y^a) < 6(3/4)^h, \quad W_2(Z_h, Z) < 7(3/4)^h. \quad (31)$$

Indeed the initial error is below $1 + 1/4 < 2$ and $2 + \sum_{j \geq 0} 2^{-j/2} < 6$. Moreover $\mathbb{E}|Y^a| < 1/4$ and $\mathbb{E}|Z| < 1/2$, as checked from the variances.

For $k = 0, \dots, 31$ let $B_k = \mathbb{E}(Y^0)^k/k!$ and $C_k = \mathbb{E}(Y^1)^k/k!$. Set $B_0 = C_0 = 1$, $B_1 = C_1 = 0$. At order $k \geq 2$, temporarily set the unknown coefficients to zero and let U_k, V_k be the coefficients of z^k in

$$\{pe^{-qz}B(z) + qe^{pz}C(z)\}^2, \quad B(z)^2.$$

With $u_k = (-1/\rho)^k$ and $w_k = 2u_k$, the exact solution is

$$B_k = \frac{u_k U_k + w_k q u_k V_k}{1 - w_k p - w_k^2 q}, \quad C_k = w_k B_k + u_k V_k.$$

The verifier encloses every denominator above $1/2$. Since $\mathbb{E}e^{2|Y^a|} \leq 2e^{2/3} < 4$, the Taylor remainder at iu is at most $4(|u|/2)^{32}$. Evaluate the degree-31 polynomials at $u = x/(-\rho)^{24}$ and apply 24 exact transformations (14). For $|x| \leq 2048$, $|u| < 1/1000$, and the propagated error is at most

$$2 \cdot 2^{24} \frac{4}{2000^{32}} < E := 2^{-300}.$$

Call the resulting entire conditional approximants $P_a(x)$ and their mixture $P(x)$. Their errors from ψ_a and ϕ are at most E on the entire real integration range.

6.2 A complete annulus and both finite-height tails

Divide $[64, 128]$ into 256 cells of width $1/4$. At a midpoint x_c put $u_c = -x_c/\rho$. The full cell is covered by the two bounds

$$\left(|P(u_c)| + E + \frac{1}{2} \frac{1/8}{48/25} \right)^2, \quad \left(|P_0(u_c)| + E + \frac{1}{4} \frac{1/8}{48/25} \right)^2. \quad (32)$$

The padding uses the first absolute moments before the final square. Every cell bound is strictly below $1/8$. By (31), for every $k \geq 128$ this transfers to

$$Q_k(u/a_k) < U := 1/4 \quad (64 \leq |u| \leq 128),$$

because $1/8 + 6 \cdot 128(3/4)^{128} < 1/4$.

Take the Fourier cutoff $T = 2048$. On a shell $64\rho^n \leq |u| \leq 64\rho^{n+1}$, the limiting modulus is at most U^{2^n} . The same bound holds at finite height whenever $|u/a_h| \leq 64/|a_{128}|$, by (19). For $|u| \geq T$ the shell index is at least five, since $\rho < 2$. For $0 \leq j \leq 3$, the corresponding tail integrals, including $1/\pi$, are bounded by

$$B_j^{\text{tail}} = 2^{12(j+1)+1-64}. \quad (33)$$

To check this, each shell endpoint is at most $128 \cdot 2^n$. Sum from $n = 5$: the first bound is $4096^{j+1}U^{32}$ and the successive ratio is at most $2^{j+1}U^{32} < 1/2$.

For the remaining physical frequencies use the seed's adjacent atoms. Each conditional seed law has two adjacent masses greater than $(1 + \lambda)^{-14} > 2^{-224}$, so $Q_0(v) \leq \exp(-2^{-452}v^2)$ for $|v| \leq \pi$. Since $|a_{128}| < 2^{128}$, the high-frequency part has $|v| > 2^{-122}$, and hence its modulus is at most

$$\exp[-2^{h-696}].$$

The integration range is shorter than 2^{h+2} . For every $h \geq 1024$, its integrals against $|u|^j/\pi$, $j \leq 3$, are less than 2^{-390} : it suffices that $2^{h-696} > 4h + 400$, which holds at 1024 and is preserved by induction. This explicitly controls the finite-height lattice tail.

6.3 Validated curvature and the all-height conclusion

Acb validated integration evaluates, for $j = 0, 1, 2$,

$$\frac{1}{\pi} \operatorname{Re} \int_0^{2048} (-iu)^j P(u) e^{-ix_* u} du,$$

using 128 segments of length 16 and absolute/relative tolerances 2^{-80} . Adding $ET^{j+1} + B_j^{\text{tail}}$ proves the exact rational enclosures

$$0.045 < f(x_*) < 0.046, \quad -0.088 < f'(x_*) < -0.086, \quad 24.9 < f''(x_*) < 25. \quad (34)$$

In particular $W(x_*) := f(x_*)f''(x_*) - f'(x_*)^2 > 1$. The central range and (33) give $\|f^{(j)}\|_\infty < 2^{11(j+1)}$ for $j \leq 3$, so $\|W'\|_\infty < 2^{56}$. Throughout J , these bounds imply

$$W > 0.99, \quad 0.044 < f < 0.047, \quad |f'| < 0.09, \quad |f''| < 26. \quad (35)$$

For every $h \geq 1024$, combining (31) and both tails bounds the three uniform finite-height errors by

$$R_0 = 2^{-49}, \quad R_1 = 2^{-37}, \quad R_2 = 2^{-25}. \quad (36)$$

Specifically, for derivative order j a sufficient bound is $7(3/4)^h T^{j+2} + 2B_j^{\text{tail}} + 2^{-390}$. The verifier checks it at $h = 1024$; it decreases thereafter.

Consequently, on J , $0.043 < g_h < 0.048$, $|g_h''| < 27$, and $|g_h'| + |f'| < 0.2$. The curvature numerator obeys

$$g_h g_h'' - (g_h')^2 > 0.99 - 27R_0 - 0.047R_2 - 0.2R_1 > 0.98.$$

Thus

$$(\log g_h)'' > 0.98/(0.048)^2 > 400.$$

Equations (18) and (24) prove (29) for every selected node.

The count is at least $2d|a_h| - 3 \geq d|a_h| \geq 2^{-82}\rho^h$: $2^{-82}(48/25)^{1024} > 4$. The exact rational inequality $(48/25)^{50} > 2^{47}$ and $N_h < 16 \cdot 2^h$ prove (27). Finally (25), $D_0 > 0$, $\mu_0^0 > 6$ and $|a_0| < 1$ give, for $|x| < 1$ and every $h \geq 1024$,

$$0 < \frac{m_h + a_h x}{2^h} < \frac{15}{2} + 4(25/26)^h < \frac{99}{100}(23/3 - 2^{-h}) \leq \frac{99}{100} \frac{\alpha_h}{2^h}.$$

For the first upper bound, the seed mixture is at most seven, and

$$\left| 2^{-h} \sum_{j < h} 2^j (b_j - \pi_*) \right| \leq \frac{q - \pi_*}{2q - 1} q^h < q^h.$$

The last numerical comparison holds at 1024 and improves with h . This proves (28) and completes the finite theorem. $\square$

7 Why exponent one still permits zero density

We include an independent finite argument, so that this distinction does not depend on the behavior of the constructed family. Fix a degree bound $\Delta \geq 3$, and put $b = \Delta - 1$. Consider a run $u, \dots, v$ of length L in an n -vertex tree. Write $d = L + 1$ and $c = (u + v)/2$. Choose the unique activity with $\mathbb{E}X = c$. Strict log-convexity in the window $[u - 1, v + 1]$ implies ordinary convexity of its tilted probabilities. The exact identity

$$\begin{aligned} & \sum_{j=0}^d \left((j - d/2)^2 - \frac{d(d+2)}{12} \right) p_{u-1+j} \\ &= \sum_{k=1}^{d-1} \frac{k(k+1)(d-k)(d-k+1)}{12} (p_{u-2+k} - 2p_{u-1+k} + p_{u+k}) \geq 0 \end{aligned}$$

follows by collecting coefficients. Outside the window the squared distance from c exceeds $d(d+2)/12$. Hence

$$\text{Var } X \geq d(d+2)/12 > d^2/12. \quad (37)$$

Gibbs entropy gives $(\alpha - c) \log \lambda \leq n \log 2$; since $\alpha - c \geq d/2$, $\lambda \leq 2^{2n/d}$.

Root the tree at a vertex u . On a directed edge to a child z , $\mathbb{E}(S_z | S_{\text{parent}}) = q_z(1 - S_{\text{parent}})$, where $q_z \leq q_* := \lambda/(1 + \lambda)$. Conditional independence on tree paths gives $|\text{Cov}(S_u, S_v)| \leq q_*^{\text{dist}(u,v)}/4$. A radius- R ball has at most $3b^R$ vertices, and therefore

$$\text{Var } X \leq \frac{1}{4}(3nb^R + n^2q_*^R). \quad (38)$$

For an integer $m \geq 1$ set

$$\Lambda = 4^m, \quad K = \lceil \log_2(12m^2) \rceil, \quad R = (\Lambda + 1)K, \quad n_m = 36m^2b^R.$$

If $d \geq n/m$, then $\lambda \leq \Lambda$ and $q_*^R < 2^{-K} \leq 1/(12m^2)$, since $(1 + 1/\Lambda)^{\Lambda+1} > 2$. For $n \geq n_m$, (38) contradicts (37). For $N \geq mn_m$, trees with $n < n_m$ trivially have runs shorter than N/m , and those with $n \geq n_m$ satisfy $L < n/m \leq N/m$. Thus $M_\Delta(N) < N/m$ for every $N \geq mn_m$. Letting $m \rightarrow \infty$ proves

$$\boxed{M_\Delta(N) = o(N) \quad \text{for every fixed } \Delta.}$$

Together with Theorem 1, this gives exponent one and zero density for every fixed degree bound at least three. $\square$

8 Reproduction and scope of the certificate

Run

```
python -m pip install python-flint==0.9.0
python verify_subcubic.py
```

The verifier writes `subcubic_certificate.json`, with its source SHA-256, the algebraic root interval, conditional seed polynomials, Fourier bounds, derivative enclosures and all check names. The executed run passed **733 checks**, including all 256 complete annulus cells and 384 validated integration segments. The certificate does not enumerate the enormous trees at height 1024: the displayed recurrences and all-height estimates are the proof that transfers the finite checks to every height thereafter.

The general exponent-one theorem uses the analytically proved valley and uniform local limit theorem. Its J_t, ζ_t and final starting height are not numerically tabulated for every t . This is an existence theorem with full quantifiers, not a claimed uniform numerical certificate for all ε . Theorem 5 is the separate fully quantified instance; it has no unspecified small parameter or unevaluated starting height.

Earlier work established multiple failures [2], consecutive blocks of several fixed lengths [3], and unbounded runs with reproducibility archives [4]. Most directly, the indexed abstract of Christopher [1] already states near-linear blocks in subcubic trees using terminal paths. The present exponent statement must be read with that attribution. No conclusion here settles an independence-sequence unimodality problem: a decreasing coefficient sequence can still fail log-concavity.

References

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